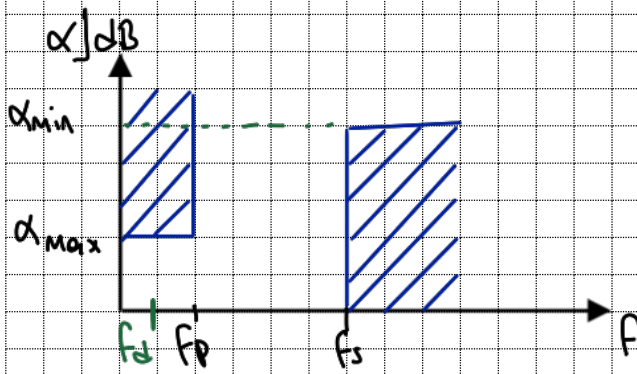


Laboratorio 1



$$\alpha_{\max} = 1 \text{ dB}$$

$$f_p = 5 \text{ kHz}$$

$$D(0) = 100 \mu\text{s}$$

$$D_n(0) = 1$$

$$\varepsilon = 2\%$$

$$f = 3 \text{ kHz}$$

$$\omega_p = \frac{D_n(0)}{D(0)} = \frac{1}{100 \mu\text{s}} = 10 \cdot 10^3 \text{ rad/s}$$

$$\omega_p = \frac{1}{100 \mu\text{s}} = 10 \text{ krad/s}$$

Normalizo con $\omega_p = 10 \text{ krad/s}$

$$\Omega_p = \frac{\omega_d}{\omega_p} = \frac{10 \text{ krad/s}}{10 \text{ krad/s}} = 1$$

$$\Omega_p = \frac{2\pi \cdot 5 \cdot 10^3 \text{ Hz}}{10 \cdot 10^3 \text{ rad/s}} = \pi$$

$$\Omega_{d_{\max}} = \frac{2\pi \cdot 3 \cdot 10^3 \text{ Hz}}{10 \cdot 10^3 \text{ rad/s}} = 0,6\pi \approx 1,885$$

Retardo de grupo

$$D(\omega) = -\frac{d\phi(\omega)}{d\omega} = \sum_{i=1}^n \frac{\theta_i}{\theta_i^2 + (\omega - \omega_i)^2}$$

con $n = 3$

$$T(s) = \frac{2,322}{s + 2,322} \cdot \frac{2,542^2}{s^2 + \frac{2,542}{0,691}s + 2,542^2}$$

$$\omega_{01} = \sqrt{2,322} \approx 1,524$$

$$\omega_{02} = 2,542$$

$$BW = \frac{\omega_0}{Q} = \frac{2,542}{0,691} \Rightarrow BW = 3,678, Q = 0,691$$

$$BW = 2\theta_i \Rightarrow \theta_i = \frac{BW}{2} \Rightarrow \theta_i = 1,839$$

$$\omega_0^2 = \theta_i^2 + \omega_i^2 \Rightarrow \omega_i = \sqrt{\omega_0^2 - \theta_i^2} \Rightarrow \omega_i = 1,755$$

$$D(\omega) = \frac{2,322}{2,322^2 + \omega^2} + \frac{1,839}{1,839^2 + (\omega - 1,755)^2} + \frac{1,839}{1,839^2 + (\omega + 1,755)^2}$$

$$D(0,6) = \frac{2,322}{2,322^2 + 0,6^2} + \frac{1,839}{1,839^2 + (0,6 - 1,755)^2} + \frac{1,839}{1,839^2 + (0,6 + 1,755)^2}$$

$$D(0,6) = \frac{2,322}{8,994} + \frac{1,839}{3,398} + \frac{1,839}{16,631} = 0,9113$$

$$\varepsilon\% = \frac{D(1) - D(0,6)}{D(1)} = \frac{1 - 0,9113}{1} \cdot 100\% = 8,87\% \text{ No cumple}$$

$$n = 4$$

$$T(s) = \frac{s^2 + \frac{3,023}{0,5219}s + 3,023^2}{s^2 + \frac{3,389}{0,8055}s + 3,389^2}$$

$$T(s) = -\frac{1}{R_1 R_3 C^2} \cdot \frac{1}{s^2 + s \cdot \frac{1}{R_2 C} + \left(\frac{1}{R_3 C}\right)^2}$$

$$\omega_0 = \frac{1}{R_3 C}$$

$$k = -\frac{R_3}{R_1}$$

$$Q = \frac{R_2}{R_3}$$

$$T(s) = \frac{3,023^2}{s^2 + \frac{3,023}{0,5219} s + 3,023^2} \cdot \frac{3,389^2}{s^2 + \frac{3,389}{0,8055} s + 3,389^2}$$

$$Q_1 = 0,5219 = \frac{R_{2,1}}{R_{3,1}} \Rightarrow R_{2,1} = 0,5219 R_{3,1}$$

$$\omega_1 = 3,023 = \frac{1}{R_{3,1} C} \Rightarrow R_{3,1} = \frac{1}{3,023 C}$$

$$|k_1| = \frac{R_{3,1}}{R_{1,1}}, k=1 \Rightarrow R_{1,1} = R_{3,1}$$

$$Q_2 = 0,8055 = \frac{R_{2,2}}{R_{3,2}} \Rightarrow R_{2,2} = 0,8055 R_{3,2}$$

$$\omega_2 = 3,389 = \frac{1}{R_{3,2} C} \Rightarrow R_{3,2} = \frac{1}{3,389 C}$$

$$|k_2| = \frac{R_{3,2}}{R_{1,2}} = 2 \Rightarrow R_{1,2} = \frac{1}{2} R_{3,2}$$

$$|T(s=j\omega)| = \frac{3,023^2}{s^2 + \frac{3,023}{0,5219} s + 3,023^2} \cdot \frac{3,389^2}{s^2 + \frac{3,389}{0,8055} s + 3,389^2}$$

$$|T(s=j\pi)| = \frac{3,023^2}{\sqrt{(3,023^2 - \pi^2)^2 + (5,79\pi)^2}} \cdot \frac{3,389^2}{\sqrt{(3,389^2 - \pi^2)^2 + (4,2\pi)^2}}$$

$$\frac{3,023^2}{18,2} \cdot \frac{3,389^2}{13,29} \cdot \frac{105}{241,87}$$

$$|T(s=j\pi)| = 0,43$$

$$|T(s=j\pi)|_{dB} = 20 \log(0,43) = -7,24 \text{ dB}$$

→

$$\Rightarrow \alpha \cdot \frac{1}{T} = 7,21 \text{ dB}$$

deseamos

$$1 \text{ dB}$$

$$\Rightarrow 7,21 - x = 1 \text{ dB} \Rightarrow$$

$$x = 7,21 - 1 = 6,21 \text{ dB}$$

Como obtenemos 7,21 dB ~~de~~ tener una ganancia de 6,21 dB $\rightarrow$

$$6,21 \text{ dB} = 20 \log K$$

$$\Rightarrow \frac{6,21}{20} = \log K \Rightarrow K \approx 1,06$$